

Auto-Engage

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Chapter 1

Introduction

The product which is described in this document is AutoEngage, which is an Artificial Intelligence, Client engagement and Sales automation solution. Specifically, AutoEngage targets to eliminate human agents in call centers so that an organization can effectively manage many client relations with little actual contact, and at a lower cost and increased efficiency.

This document is intended for a variety of readers, including:

1. **Developers:** For comprehending what actually the system architecture, APIs or the modules to be implemented consist of.
2. **Project Managers:** For passage of time management, and coordination of development and growth of new activities.
3. **Testers:** To conduct much factual testing on the functionalities, in order to ascertain that it operates to its optimal capacity.
4. **End Users (Call Center Operators/Managers):** To know how to navigate through the system, how to deal with clients & how to go through analytics.
5. **Marketing Staff:** To attract the attention and encourage the potential client to pay attention to the peculiarities of the offered services.
6. **Documentation Writers:** To write instructions for using a product or service, assembling instructions, and technical information.

1.1 Problem Statement

It becomes extremely difficult to handle and communicate with clients especially when the firm is operating in the current era of globalization. The problems are typical for the conventional call centers, including these situated in Pakistan; it is due to these reasons that it is challenging for organizations of this type to deliver high quality services over the prolong period. These include time, culture, language and human agents who may tire, make a slip, or may just not understand what the situation requires, and this is very frequent when dealing with foreign Clients. In addition, high expenses incurred in getting and training human Agents also form another disadvantage that hinders the expansion of these companies in a cheaper manner. In most circumstances, such problems result to the erosion of profit, levels of satisfaction among Clients going down, and with interacting markets such as the US, the rate of conversion coming down.

1.2 Scope

The AutoEngage project seeks to have a fully automated call center that will replace human beings by using the phone to interact with the clients. This solution deploys Speech to Text to transcribe real time client conversation, Large language models to mimic natural language/ human responses to client conversation and Text to speech to translate these responses back to speech.

It aims to solve crucial issues that outdated large call center faces including high operating cost, problems with management of overseas clients and problems with scalability. AutoEngage automates these processes and hence provides all the service without fail and without the need to consult a time difference chart regarding the clients located in different countries. One of the system's advantages is the presence of an analytics panel that tracks the progress and optimizes it. The first goal is to get more clients in an hour, increase the percentage of correct orders and solve the problem of being too formal. AutoEngage will be mainly used for keeping relationships with the customers and answering their product inquiries using script-based sales conversations, the aim being to provide a voice as close to human as possible in order to reassure customers and increase their confidence. Still, it is worth mentioning that, certain enhancements including voice cloning, deepfake, and multiple languages are not within the scope of this work.

In sum, AutoEngage targets to offer a proven and growing all-in-one software solution for call centres that will improve clients' satisfaction, optimise work, and contain expenses, thus confirming the instrument's effectiveness as a large-scale solving tool for call centre activity..

1.3 Modules

1.3.1 Speech to Text Model and Basic UI

This module focuses on developing a Speech-to-Text (STT) model and integrating it with a user interface, allowing voice input to be converted into text.

1. STT Model construction and optimization.
2. Implement a Desktop/Web App with links/Options for voice entry options including a Microphone button.
3. Integrate the Desktop/Web App mic button with the developed STT Model to capture and transcribe voice input.

1.3.2 Intent Classification Development

This idea deals with the development of the intent classification in particular, which is aimed at recognizing human's intents or particular goals.

1. Develop the Intent Classification Model.
2. Prepare Data to train the intent classification model
3. Train and achieve good accuracy for the classification of user intent to predefined classes

1.3.3 LLM Script Adaptability and Fine-Tuning

This module is focused on choosing of the Large Language Model and its further tuning to fit the conversation of the call-centre and provide the most suitable variant of the respond.

1. Select the best LLM based on base accuracy and performance for the call center use case.
2. Select only one LLM out of the trained LLMs and fine-tune the selected LLM with the help of a call center dataset in order to enhance the degree of correctness of the response.
3. To obtain flexibility in producing scripts, regarding their language and construction, apply the intent classification model with the use of an RAG approach.

1.3.4 Human-Like Speech Synthesis

Determining in advance that the computer speaking should sound like a human being is a criterion of human-like synthesis of speech by the computer. This particular module compels the development of a TTS synthesis model that would enable Avise customers get real like human voice.

1. Text-to-Speech Synthesis Model should be made performant to provide accurate results and improve its capability.
2. Make sure that the model is capable of generating the human such as male and female type voice.

1.3.5 Call Center Dashboard and Application Integration

This module focuses on building a dashboard for call center statistics and integrating it with voice and backend services.

1. Develop a Desktop/Web App with a dashboard displaying call center statistics.
2. Incorporate an in-built voice recorder for recording conversations.
3. Continuously update text in the user interface as voice is transmitted.
4. Integrate with external APIs used by the call center to initiate calls.
5. Update the UI and statistics based on call status.
6. Connect the application with a backend PostgreSQL database for storing call center statistics and customer information.

1.3.6 Pipeline Integration and Testing

This module integrates all components into a single system and ensures functionality through testing.

1. Integrate the entire pipeline with the Desktop/Web Application.
2. Ensure that all pipeline endpoints are correctly connected to the Application.
3. Conduct automated UI tests to verify functionality.
4. Perform manual testing for various call scenarios made through the Application.

1.4 User Classes and Characteristics

User Class	Description
Call Centers	Any organization with a massive call centre or customer service team depending on fielding a significant number of customer inquiries. They require a solution which is able to handle large number of clients domestically and internationally overcoming barriers such as those exhibited by time difference. AutoEngage will assist them in cutting their operating expenses because part of the pipe can be handled through marketing automation while still serving clients optimally well.
Overseas Clients	Business houses that have international clients this time zone difference is one of the most important factor to consider. AutoEngage also enables overseas clients to receive responses instantly and appropriate in spite of delay due to time difference or limited manpower. It will guarantee uninterrupted services to overseas markets and hence enhance the satisfaction level of clients.
Customer Support Teams	Some teams may be designated responsibility of managing customer satisfaction and handling complaints. These users will be able to set AutoEngage to automatically perform some of the common interactions while retaining human agent's input for most complicated transformations. The AI will also be used for generating intelligence from analytics that can be used in customer relations enhancement.
Sales Teams	Sales departments that want to introduce an initial client contact and evaluation robotization. The AutoEngage will be built with explicit sales scripts to help communicate with potential consumers, respond to simple questions, and develop as many leads for salespeople as possible. This will optimise the number of clients seen in one hour and boost the number of conversions resulting from quick responses.

Table 1.1: User Classes and their Characteristics

Chapter 2

Project Requirements

2.1 Event Response Table

For AutoEngage, an Event-Response Table is suitable.

Event	System Response
Call center initiates a call to the client	The system dials the client's number, establishes a connection, and introduces the agent as calling on behalf of the company.
Client picks up the call	The system plays the introductory script and provides basic information (e.g., the agent's name, company).
Client asks about the current package	The system uses pre-loaded data to inform the client about their current package details and benefits.
Client is offered a new insurance package	The system presents an upsell offer, explaining the benefits of subscribing to an additional or upgraded package.
Client accepts the new package	The system confirms the upgrade and marks the call successful as it will be transferred to the live agent
Client declines the offer	The system logs the client's response, thanks them for their time, and ends the call and marks the status as accordingly i.e (Not Interested, Do not call)
Client deviates from the call purpose	Based on the client's response, the system will address that response and bring the pitch of the sale back on track.
Client requests more information	The AI Agent gives out more information based on what the client is asking about

Table 2.1: Event Response Table

2.2 Functional Requirements

In this section, the functional requirements for the AutoEngage system are presented and discussed. The requirements are summarized according to feature and the specific identified functional requirement associated with each module.

2.2.1 Speech to Text Model and Basic UI

Following are the requirements for this module:

1. **FR1:** The system also requires interfacing with an existing Speech-to-Text (STT) model that continuously transcribes actual speech of the clients in various situations into writing.
2. **FR2:** The system must have a possibility to be run on the desktop and/or via the web interface, there should be a microphone button for user interaction.
3. **FR3:** The microphone button has to be added to the STT model for the purpose of voice recognition in real time and speech recognition in real time

2.2.2 Intent Classification Development

Following are the requirements for this module:

1. **FR1:** It is a necessity to incorporate an intent classification model into the system to categorise the client input into pre-ordained categories.
2. **FR2:** The intent classification model also needs to assign the correct intent for each user's request into the defined categories

2.2.3 LLM Script Adaptability and Fine-Tuning

Following are the requirements for this module:

1. **FR1:** The system must consist of a Large Language Model (LLM) based on base Suitability, precision, and throughput and for call center use cases suitability.
2. **FR2:** The selected LLM must be fine-tuned with call center type data to increase response accuracy.

3. **FR3:** The system has to integrate the proposed LLM model with Retrieval-Augmented Generation. process, the script generation in other words is intended to create a rather elastic RAG for a dialogical. interactions with the client.

2.2.4 Human-Like Speech Synthesis

Following are the requirements for this module:

1. **FR1:** It has to contain the Text-to-Speech (TTS) synthesis model that generates human-like speech.
2. **FR2:** The TTS model must make use of male and female voice too that sound natural.

2.2.5 Call Center Dashboard and Application Integration

Following are the requirements for this module:

1. **FR1:** The system must offer a solution delivered through a desktop/web application, complete with a dashboard designed to display the play call center numbers (e.g. number of calls, average length of a call, customer satisfaction).
2. **FR5:** This application must integrate with an associated backend PostgreSQL database where the population includes center statistics and client information.

2.2.6 Pipeline Integration and Testing

Following are the requirements for this module:

1. **FR1:** The whole process should be linked to a desktop/web application thus that each of the parts is integrated in the best manner possible.
2. **FR2:** There has to be an automated UI testing to ensure that the system is operating properly.
3. **FR3:** The system has to be manual tested for different call scenarios in order to verify their accuracy and reliability

2.3 Non-Functional Requirements

The feature/model that follows defines other requirements that autodialing systems like AutoEngage come with. These where quality requirements are precise, tangible and executable, to assure the corresponding system. This involves the aspect of rating according to expectations on performance, reliability, usability as well as security.

2.3.1 Reliability

The system must ensure a high level of reliability to maintain continuous call center operations.

- **R-1:** The system should have an uptime of 99.9 percent with no more than 5 minutes of downtime per month.
- **R-2:** Should there be system failure, then all the recovery processes must be carried out automatically and the Operating system should comeback online within one hour of breakdown.

2.3.2 Usability

The system must be user-friendly and accessible for both call center operators and administrators.

- **USE-1:** Operators managing the AI should have access to a clear, intuitive dashboard for viewing ongoing calls, metrics, and real-time issues.

2.3.3 Performance

AutoEngage must meet performance benchmarks to handle large-scale operations.

- **PER-1:** The STT model should maintain an accuracy rate of 90 percent or higher.
- **PER-2:** The dashboard must load and display real-time call analytics data within 10 seconds of request.
- **PER-3:** The TTS model should generate speech that is rated at least 4/5 in terms of naturalness by users.
- **PER-4:** Intent classification model should be able to classify the user's intent correctly 95 percent of the time.

- **PER-5:** The fine-tuned LLM should be able to generate responses in line with the user intent or the script with a 95 percent accuracy.

2.3.4 Security

AutoEngage must be comply with the following security requirements.

- **S-1:** Only authorized users should access system features, and all data should be securely stored.

2.4 Domain Model

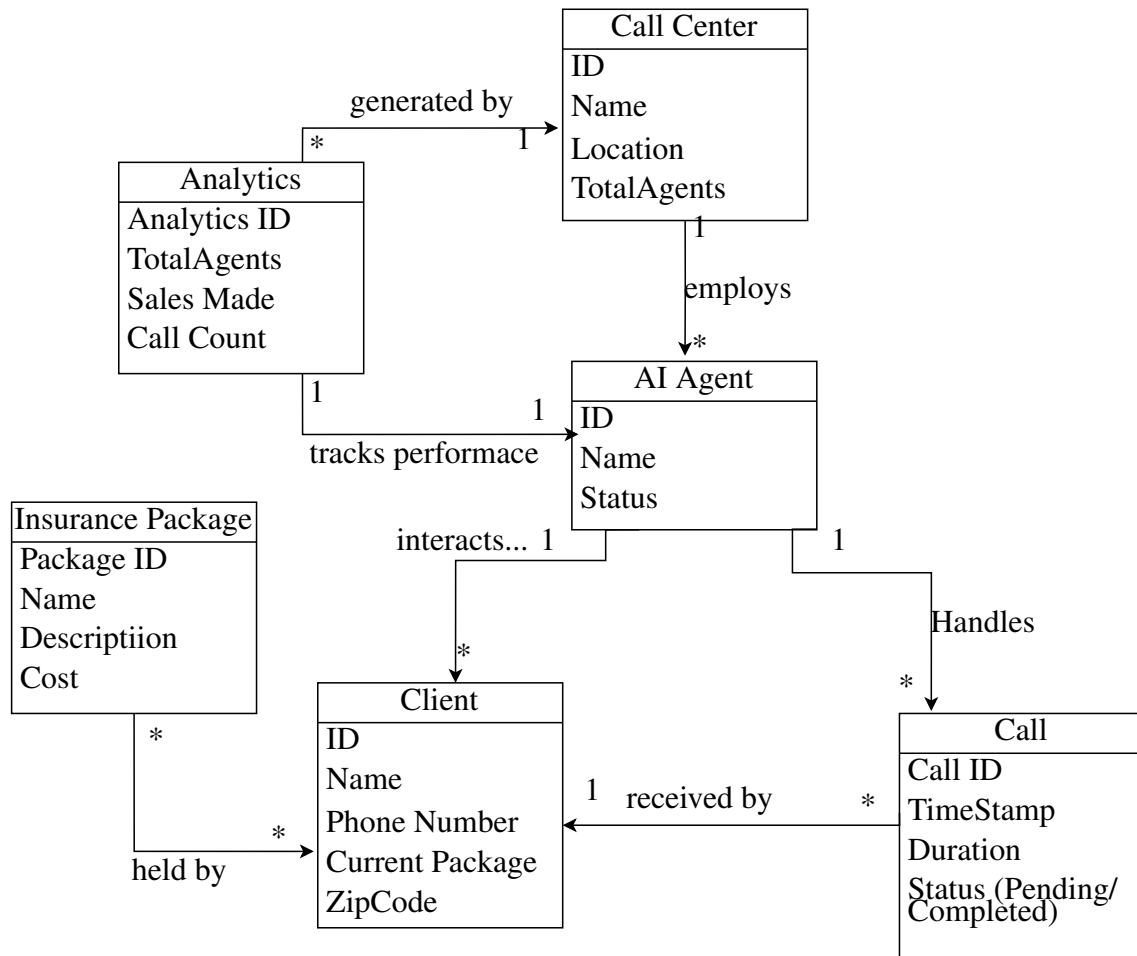


Figure 2.1: Domain Model for AutoEngage

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 General Description

The AutoEngage system is an AI-powered client engagement and sales automation platform designed to streamline and enhance call center operations. The primary goal of AutoEngage is to replace human agents with AI-driven interactions, enabling businesses to handle a large volume of client interactions with minimal human intervention. This automation not only reduces operational costs but also ensures scalability and consistent performance across various client interactions.

The system leverages advanced technologies such as Speech-to-Text (STT) for transcribing real-time conversations, Large Language Models (LLMs) to generate human-like responses, and Text-to-Speech (TTS) to relay these responses back to clients. AutoEngage is intended to improve client interactions by offering quick, accurate, and personalized communication. Furthermore, it features an analytics module that tracks and evaluates the performance of agents, calls, and overall system operations.

3.2 Context

In the modern call center environment, companies face several challenges such as high operational costs, limitations of human agents, and the complexity of managing global clients. Traditional call centers often struggle to meet these demands due to time zone differences, human fatigue, and the cost of hiring and training agents.

AutoEngage addresses these challenges by automating interactions with AI-driven agents

that can handle a wide range of client inquiries, upsell opportunities, and service requests. The system is designed to operate 24/7, allowing businesses to manage client interactions across different time zones without additional staffing. Additionally, AutoEngage provides insights into customer behavior and agent performance through real-time analytics, enabling call centers to optimize their operations continuously.

3.3 Design Overview

The design of the AutoEngage system is modular and scalable, ensuring that it can be easily integrated into existing call center infrastructures. The system is composed of the following key modules:

- **Speech-to-Text (STT) Module:** This component transcribes the client's speech into text, allowing the system to understand and process client inquiries.
- **Intent Classification Module:** Once the client's speech is converted to text, this module classifies the intent of the client, such as a request for information, a complaint, or an interest in purchasing additional services.
- **Large Language Model (LLM) Module:** The LLM generates human-like responses based on the classified intent. It adapts the response according to predefined sales scripts or the client's inquiry.
- **Text-to-Speech (TTS) Module:** This module converts the generated text responses back into speech, which is then delivered to the client, creating a seamless and human-like interaction.
- **Call Center Dashboard:** The system includes a web-based dashboard that displays real-time call center analytics, agent performance, and customer satisfaction metrics. The dashboard is accessible to managers and supervisors to monitor system efficiency and adjust operations accordingly.
- **Analytics Module:** This module collects data on client interactions, agent performance, call outcomes, and customer feedback. It generates reports that help the call center to make data-driven decisions and improve overall efficiency.

3.4 System Workflow

The workflow of AutoEngage can be summarized as follows:

1. The system initiates a call to a client from the call center's database.
2. When the client answers, their speech is transcribed into text by the STT module.
3. The Intent Classification Module analyzes the transcribed text and determines the purpose of the call.
4. The LLM generates an appropriate response based on the client's intent and predefined scripts.
5. The response is converted back into speech by the TTS module and delivered to the client.
6. The system tracks the client's responses and decisions, offering upgrades or further assistance as needed.
7. At the end of the call, the analytics data is updated, providing real-time insights into the call's outcome and the agent's performance.

3.5 Background Information

The need for automated systems like AutoEngage has grown in recent years due to the increasing demand for cost-efficient and scalable call center solutions. Many businesses, especially those with global operations, require systems that can handle a high volume of customer inquiries without relying heavily on human agents. By integrating AI technologies into call center workflows, AutoEngage helps businesses achieve greater efficiency, reduce operational costs, and improve customer satisfaction.

AutoEngage is built on modern AI models such as OpenAI's GPT, with adaptations for the specific needs of call centers. The system's design ensures that it can handle multiple languages and client profiles, though advanced features like voice cloning and deepfake technologies are not within its current scope.

3.6 Architectural Design

In this section, we will develop a modular program structure for the AutoEngage system based on the Client-Server Architecture. This structure ensures that each part of the system focuses on a specific responsibility, while still allowing different components to collaborate effectively to achieve the overall functionality.

3.6.1 Modular Program Structure

The AutoEngage system is composed of two main components: the Client Side and the Server Side, each with its own set of subsystems. These components collaborate through well-defined interfaces to ensure smooth and efficient operation of the system.

3.6.1.1 Client Side

The Client Side is responsible for managing user interactions and presenting data in a user-friendly format. It contains two main modules:

- **User Interface (Call Center Dashboard):** This module provides the front-end interface for call center agents and managers to interact with the system. It displays real-time analytics, agent performance data, and call status updates. It also allows users to initiate or terminate calls.
- **Client Request Handling:** This module manages communication between the client and server, sending user requests (e.g., starting a call or retrieving analytics) to the server, and receiving the necessary responses (e.g., call outcomes or system health information).

3.6.1.2 Server Side

The Server Side handles the core business logic, data processing, and data storage. It consists of the following functional modules:

- **Speech-to-Text (STT):** This module transcribes client speech into text for further processing.
- **Intent Classification:** This module analyzes the transcribed text to determine the client's intent (e.g., inquiry, complaint, or service request).
- **Response Generation (LLM):** This module generates human-like responses based on predefined scripts and the client's detected intent.

- **Text-to-Speech (TTS):** This module converts the generated text responses into natural-sounding speech, allowing the system to maintain a conversational interface with the client.
- **Call Management:** This module handles the entire lifecycle of a call, from initiation to termination, tracking the duration and managing call states.
- **Analytics:** This module collects and analyzes data from completed and ongoing calls, providing insights into agent performance, client satisfaction, and system efficiency.
- **Data Storage:** This module ensures persistent storage of call data, agent performance metrics, and analytics. It interfaces with a database for secure and efficient data management.

3.6.2 Relationships Between Modules

The interaction between the Client Side and Server Side is based on a request-response model. The following high-level sequence describes the flow of data between the modules:

3.6.2.1 Client Side → Server Side

- The client sends a request (e.g., initiating a call) through the *Client Request Handling* module to the server.
- During a call, the client's speech is sent to the *Speech-to-Text* module on the server, which transcribes the speech for further processing.

3.6.2.2 Server Side → Client Side

- Once the *Speech-to-Text* module has transcribed the client's speech, the text is passed to the *Intent Classification* module, which identifies the intent behind the client's request.
- Based on the classified intent, the *Response Generation (LLM)* module generates an appropriate response.
- The *Text-to-Speech* module converts the generated text response into speech and sends it back to the client through the *Client Request Handling* module.

3.6.2.3 Server-Side Module Interactions

- The *Speech-to-Text* module transcribes audio data and passes it to the *Intent Classification* module for processing.
- The *Intent Classification* module analyzes the transcribed text to determine the client's intent and forwards this to the *Response Generation (LLM)* module.
- The *Response Generation (LLM)* module generates a dynamic response, which is then sent to the *Text-to-Speech* module for conversion into natural-sounding speech.
- The *Call Management* module oversees the call's progress, ensuring proper handling of its lifecycle and updating the *Analytics* module with relevant data (e.g., duration, agent performance, and client satisfaction).
- All data related to the call, including performance metrics, are stored securely in the *Data Storage* module for future retrieval and analysis.

3.6.3 Box and Line Diagram (Initial Design)

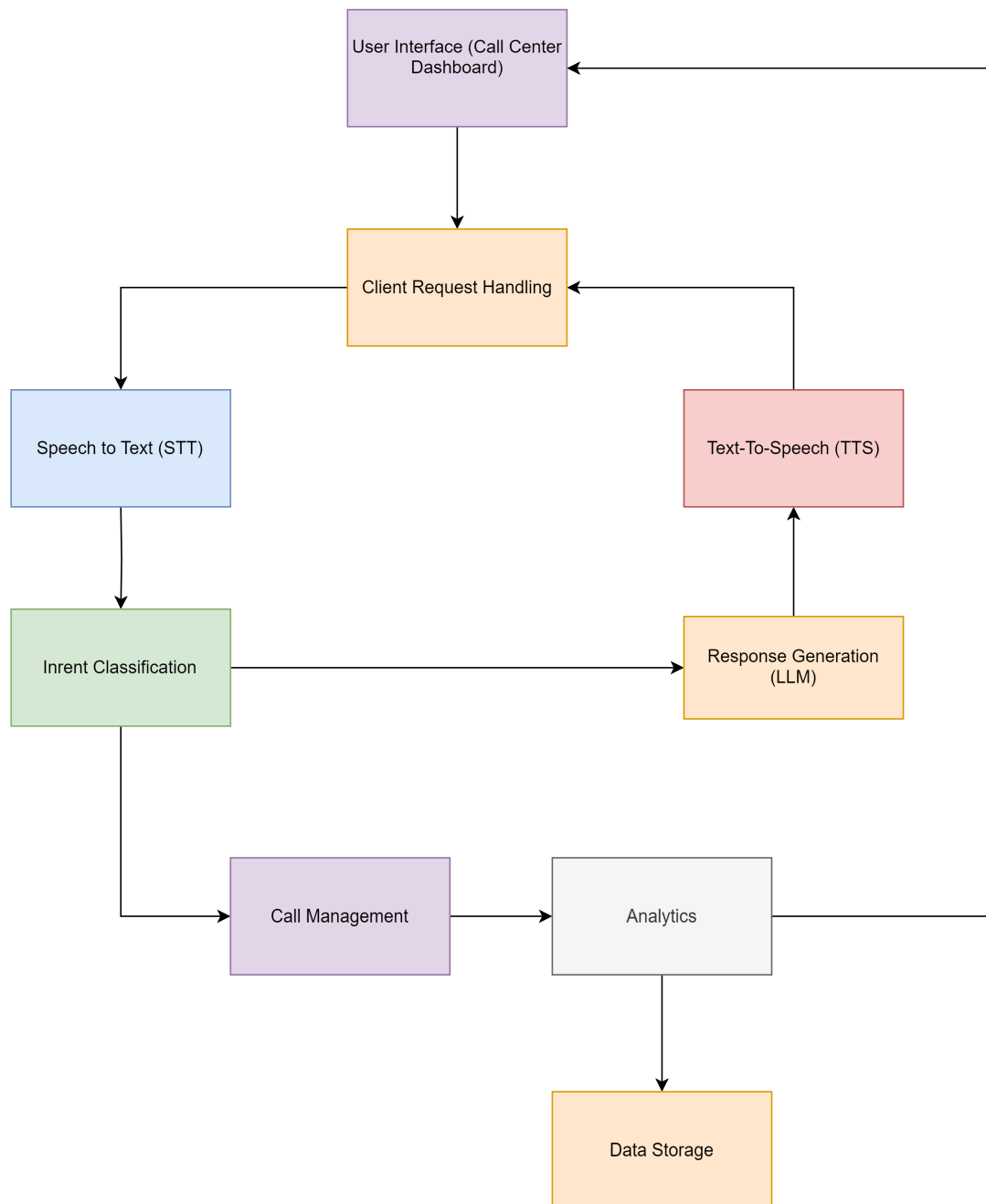


Figure 3.1: Box and Line Diagram for AutoEngage

3.6.4 Final Architectural Style: Client-Server

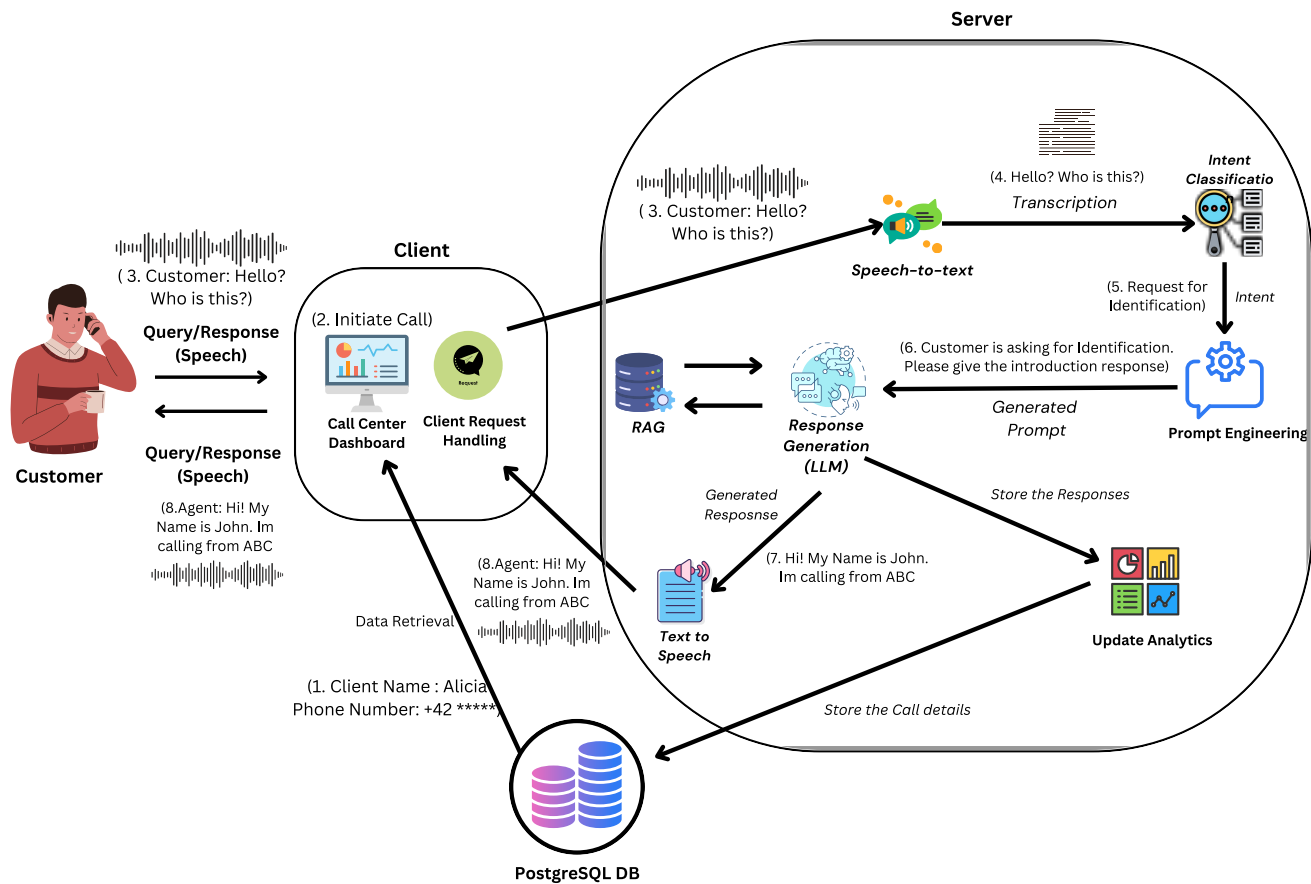


Figure 3.2: Architecture Diagram

3.7 Design Models

3.7.1 Activity Diagram

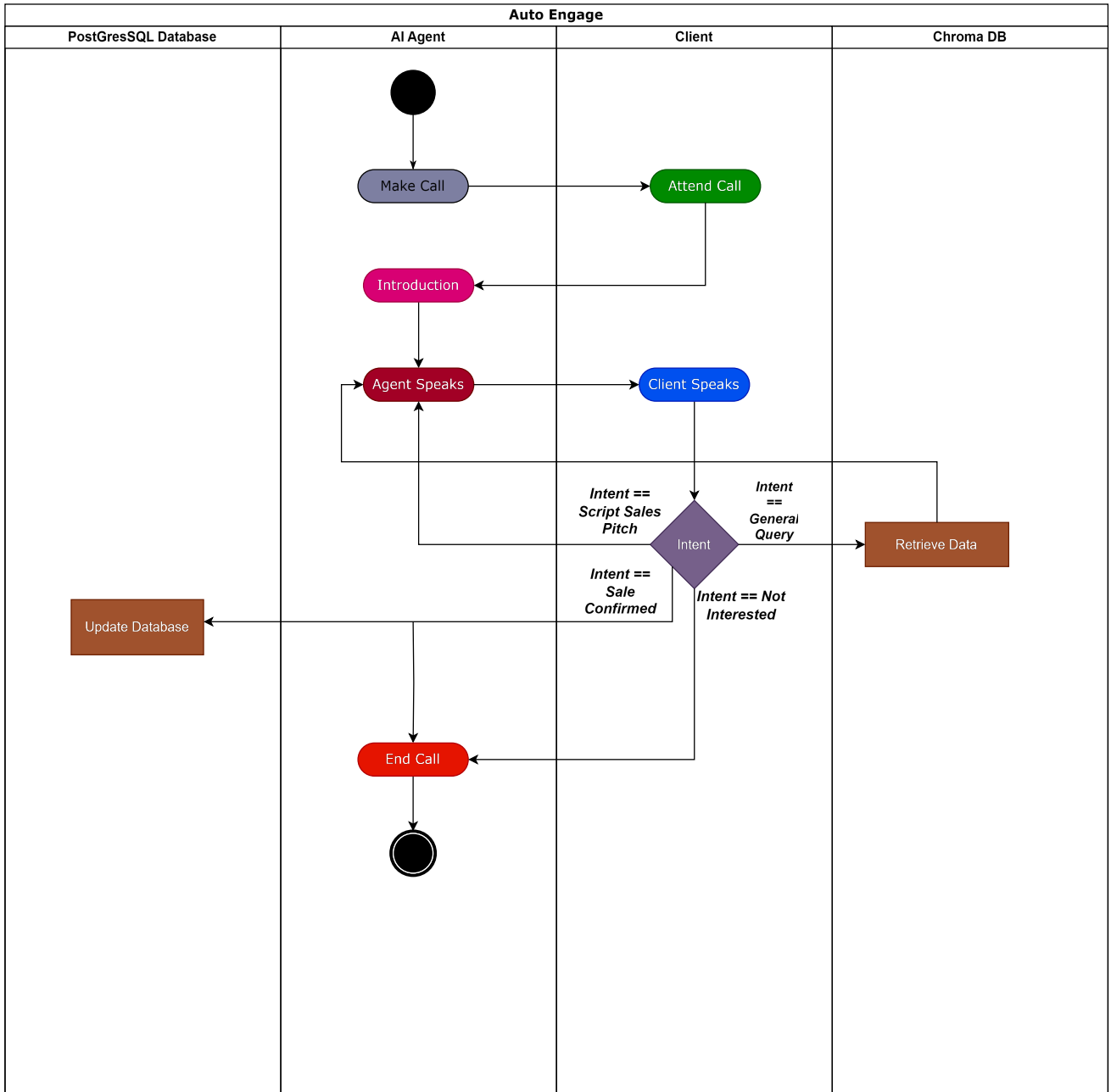


Figure 3.3: Activity Diagram

3.7.2 Data Flow Diagram

3.7.3 Level 0: Context Diagram

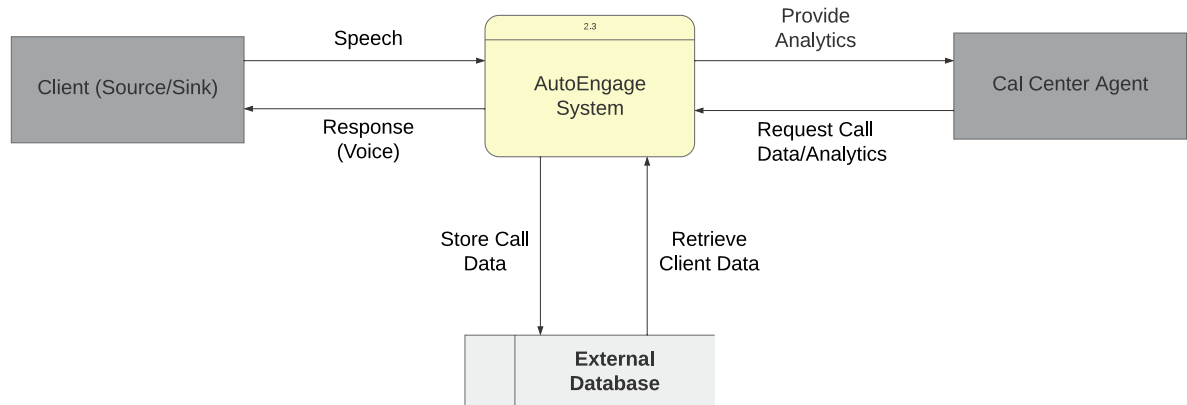


Figure 3.4: DFD Level 0

3.7.4 Level 1: Process Breakdown Diagram.

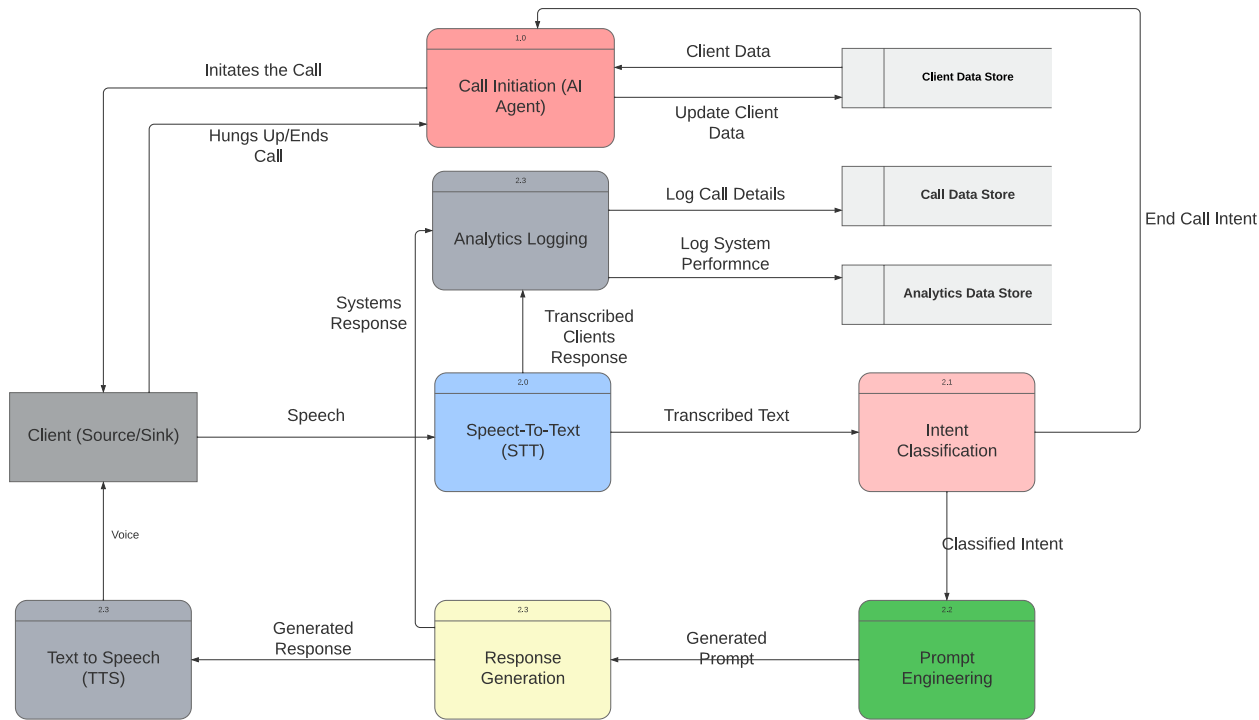


Figure 3.5: DFD Level 1

3.7.5 System-level Sequence Diagram

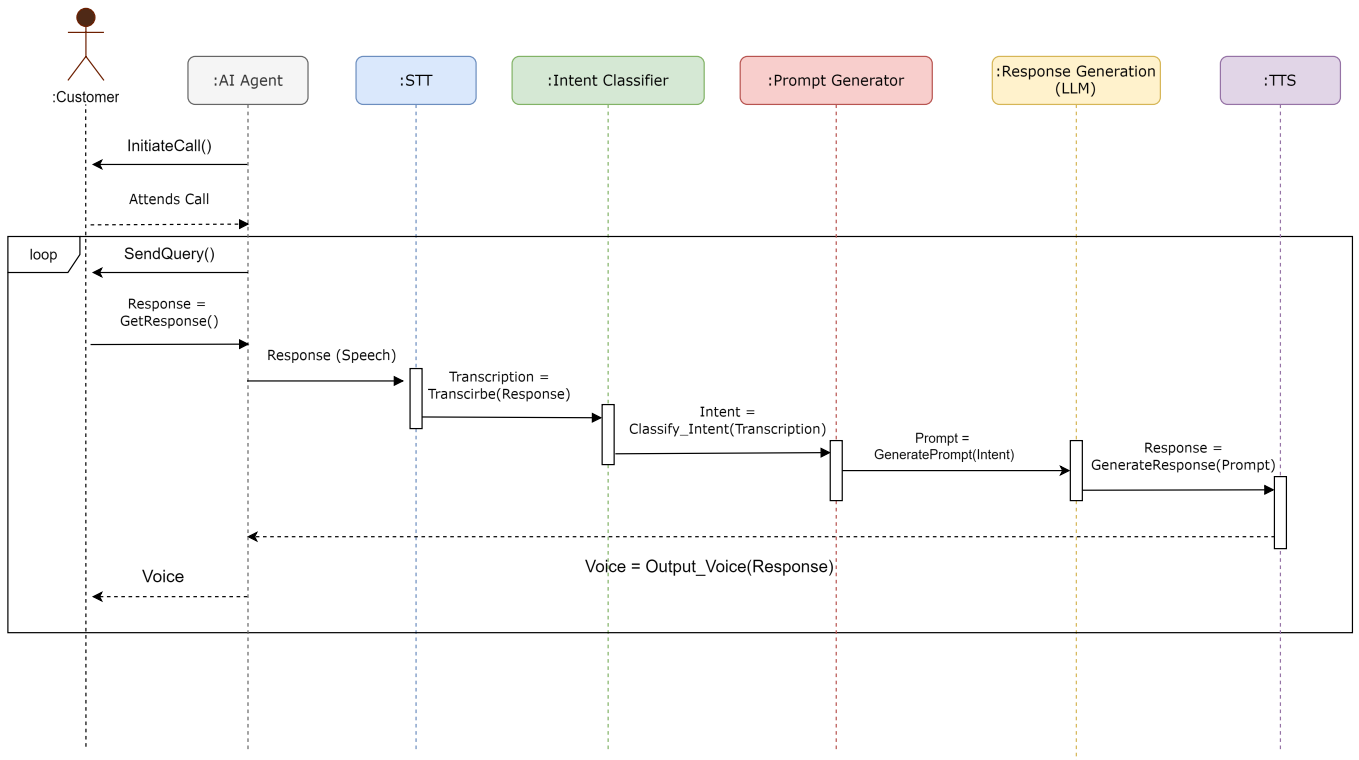


Figure 3.6: System Level Sequence Diagram

3.7.6 State Transition Diagram

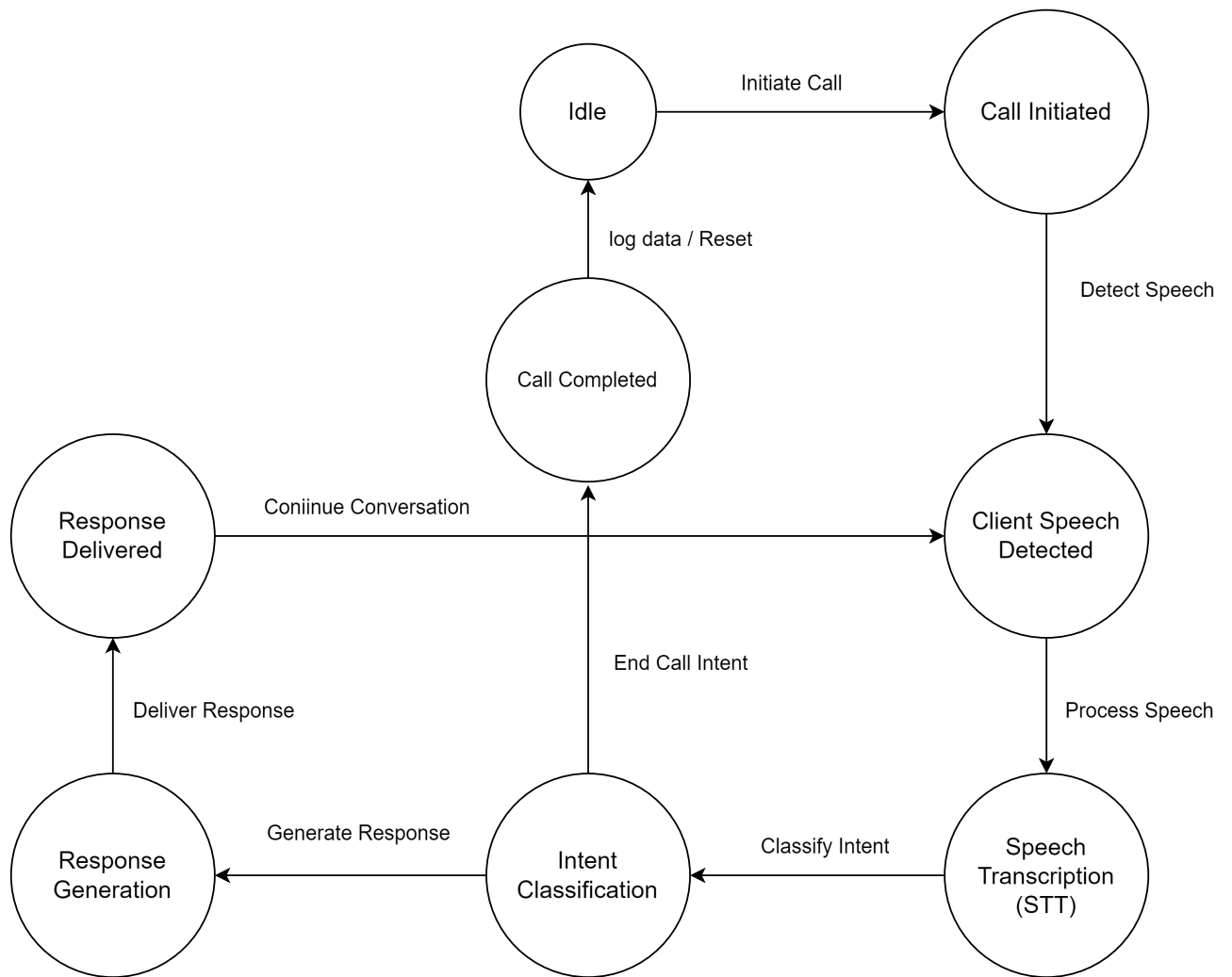


Figure 3.7: State Transition Diagram

3.8 Data Design

Data is one of the essential aspects in the AutoEngage AI-powered Call Center system and the core of all the call interaction with clients including the information about particular discussions, profiles, call history, and statistics. This section conceptualizes the content domain of the system and how the data entities are stored, processed and organized.

3.8.1 Data representation diagram: ERD

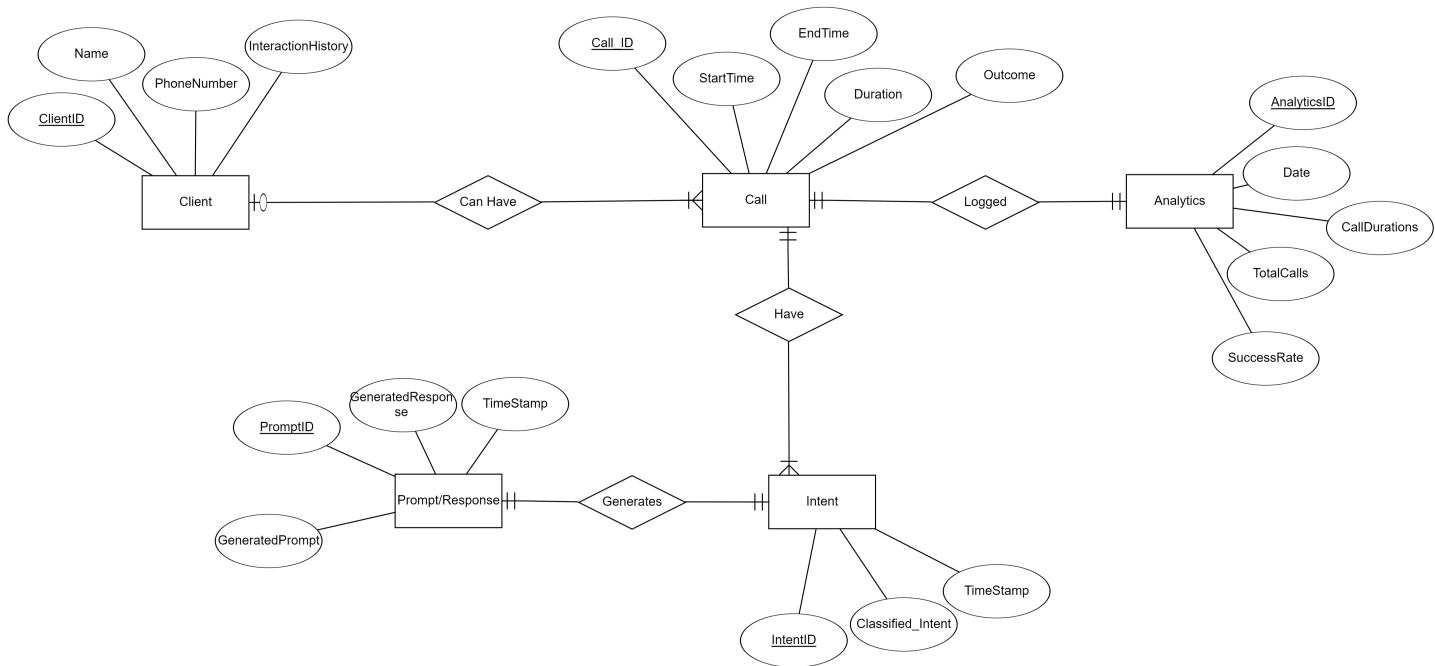


Figure 3.8: ERD Diagram

3.8.2 Key Data Structures

3.8.2.1 Client Information

Purpose: Stores information about each client who receives calls from the AI agent. This data includes the client's name, contact details, and call history.

Data Structure:

- Client ID (unique identifier)
- Name

- Phone number
- Interaction history (previous conversations, call status)

Storage: This data is stored in the *Client Data Store*.

3.8.2.2 Call Information

Purpose: Tracks the details of each call initiated by the AI agent, including the call duration, timestamp, transcription, and call outcome.

Data Structure:

- Call ID (unique identifier)
- Client ID (reference to the client data)
- Start time and end time
- Call duration
- Transcribed conversation (text from Speech-to-Text module)
- Call outcome (e.g., successful, client hung up, follow-up needed)

Storage: Stored in the *Call Data Store* for retrieval and analysis.

3.8.2.3 Intent Classification Data

Purpose: Stores the classification of client intents based on the transcribed speech. It's used for determining how to respond to the client.

Data Structure:

- Intent ID
- Client ID
- Classified intent (e.g., request for information, complaint, product interest)
- Intent timestamp (when the intent was detected)

Storage: Temporarily stored during the call and logged in the *Call Data Store* for post-call analytics.

3.8.2.4 Prompt and Response Data

Purpose: Stores the prompt generated by the Prompt Engineering module and the response generated by the LLM (Large Language Model).

Data Structure:

- Prompt ID
- Intent ID (reference to the classified intent)
- Generated prompt (text-based prompt for LLM)
- Generated response (text response provided by the LLM)
- Timestamp (when the response was generated)

Storage: Logged in the *Call Data Store* for future review.

3.8.2.5 Analytics Data

Purpose: Stores overall performance and system metrics, such as the number of successful calls, average call duration, client satisfaction, and AI system accuracy.

Data Structure:

- Analytics ID
- Date of report
- Total calls
- Successful interactions
- Average call duration
- Client satisfaction rating
- Agent performance metrics

Storage: Stored in the *Analytics Data Store*, which is accessed for performance reporting and decision-making.

3.8.3 Databases and Data Storage Items

The system includes two main databases for storing call-related and analytics data:

3.8.3.1 Client Data Store

Type: Relational database.

Purpose: Stores client profiles and interaction histories.

Structure:

- Client ID (Primary Key)
- Name
- Contact information
- Interaction history (linked to call logs)

Access: Accessed during call initiation to retrieve client details, and updated after each call.

3.8.3.2 Call Data Store

Type: Relational database.

Purpose: Stores call details, including call transcriptions, intents, and responses.

Structure:

- Call ID (Primary Key)
- Client ID (Foreign Key)
- Transcription
- Call outcome
- Call duration
- Generated responses

Access: Continuously updated during the call and after the call ends. Used for generating reports or analyzing past conversations.

3.8.3.3 Analytics Data Store

Type: Relational database (optimized for analytics).

Purpose: Stores performance metrics and call center analytics.

Structure:

- Analytics ID (Primary Key)
- Date
- Total calls
- Success rates
- Average call duration
- Performance and satisfaction metrics

Access: Updated periodically (e.g., daily, weekly), based on call center performance. Provides insight into overall system efficiency and improvements.

3.8.4 Data Flow and Processing

The system follows these general steps to process and store the data:

1. **Client Call Initiation:** The AI agent retrieves client information from the *Client Data Store* and initiates a call.
2. **Speech Processing:** The client's speech is converted to text by the *Speech-to-Text* (STT) module and stored in the *Call Data Store*.
3. **Intent Classification:** The transcribed text is classified to understand the client's intent and logged in the *Call Data Store*.
4. **Response Generation:** Based on the classified intent, a prompt is generated, and the *LLM* generates a response. The generated response is stored along with the intent in the *Call Data Store*.
5. **Analytics Logging:** After the call, key performance metrics such as call duration, client intent, and call outcome are logged in both the *Call Data Store* and the *Analytics Data Store* for future analysis.

3.8.5 Conclusion

The data design for the AutoEngage system focuses on efficient storage and processing of client and call information. Data structures are designed to ensure smooth interactions between the AI agent and clients, while the analytics component tracks and improves system performance. The system employs multiple data stores to keep track of client profiles, call histories, and performance metrics, ensuring that every interaction is logged and stored for future analysis.

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